

Swetha Vaidyanathan Sriram

Software Engineer, AI/ML

✉ vsswetha@gmail.com 📞 925-750-0169 📍 PA (Open to Relocation) 🌐 swetha-sriram 📱 swetha7502 📄 Portfolio

EDUCATION

Master of Science, Computer Science

The Pennsylvania State University, Pennsylvania

2025 – 2027 | CGPA: 4/4

Bachelor of Engineering, Computer Science and Engineering

Madras Institute of Technology, Anna University, India

2020 – 2024 | CGPA: 3.94/4

Bachelor of Science, Data Science & Programming (Hybrid)

Indian Institute of Technology, Madras, India

2020 – 2024 | CGPA: 3.73/4

SKILLS

Languages — Python, Java, C++/C, JavaScript/TypeScript, C#, Bash, SQL

Databases — PostgreSQL, MySQL, MongoDB, Redis, SQLite

Web & Backend — FastAPI, Flask, ReactJS, Vue.js, Next.js, Node.js, HTML5, CSS3, REST APIs

AI & ML — LangSmith, LangChain, LangGraph, PyTorch, ChromaDB, OpenCV, Scikit-Learn, TensorFlow

Tools & Platforms — AWS, GCP, OCI, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Celery, CI/CD, Oracle APEX

Core CS — Data Structures & Algorithms, System Design, OOP, Git

WORK EXPERIENCE

AI Solutions Intern | Penn State - Global Engagement Office

Jul 2026 – Present

- Built AI-driven optimization features using **Python**, **LLM APIs**, and **React** for a campus Event Efficiency and Sustainability Platform, covering scheduling, waste tracking, and attendee logistics
- Designed LLM-powered workflows to automate event planning decisions, reducing manual coordination across the Office

Research Assistant | Pennsylvania State University

Jan 2026 – May 2026

- Accelerated molecular dynamics simulations by 15%, analyzing 10,000+ trajectories across 50 protein complexes using **C#** and **MATLAB**.

Associate Applications Developer | Oracle Pvt. Ltd.

Aug 2024 – Aug 2025

- Enhanced an enterprise-scale release management platform (100+ users across APAC and EMEA) by **designing and deploying 2 new workflows** using **PL/SQL**, **JavaScript**, and **Oracle APEX**, reducing manual scheduling (~20% improvement).
- Designed and developed a **Jira ticket dashboard** (150+ DAUs) with real-time status visibility, reducing weekly status meetings by ~20% and accelerating ticket resolution time by 5%
- Deployed **3 Oracle SDK releases**, implementing core features, bug fixes and delivery of enterprise updates.
- Resolved critical bugs and enhancements in **Java** and developed **JUnit** tests, improving system reliability
- Analyzed and optimized **Oracle Cloud Infrastructure** (OCI) logs to identify and eliminate redundant entries.

Project Intern | Oracle Pvt. Ltd.

Jan 2024 – Jul 2024

- Automated** regression **testing flows** with Oracle Utilities Testing Accelerator (**OUTA**) and **Selenium**, boosting test coverage by ~30% and reducing manual QA effort by ~40%
- Built and deployed an energy forecasting model adopted by 10+ data science team members using **Oracle ADB**, **Python**, **Java** and **Selenium**

ACADEMIC PROJECTS

CodeBase Intel - Code Intelligence Platform 📄

- Built a polyglot tree-sitter AST parser that models cross-file dependencies in **PostgreSQL** (Supabase) with recursive CTEs and renders an interactive graph via **React Flow** + **Vite**; async ingestion offloaded to **Celery** + **Redis**
- Enabled semantic code search and **RAG**-based Q&A using **ChromaDB** + **OpenAI**; deployed to **GCP Cloud Run** with multi-stage **Docker** builds, **Terraform IaC**, and **GitHub Actions CI/CD**

ConflictMind - AI Agent that Remembers You 📄

- Engineered a 3-step debate pipeline using **Google ADK** + **Gemini 2.5 Flash (Vertex AI)** that detects contradictory memories via vector similarity + LLM classification, resolves them through structured debates, and persists unified memory to MongoDB Atlas
- Integrated a **MongoDB MCP** server for dynamic agent memory I/O; containerized with **Docker** and deployed to **GCP Cloud Run** via **Flask** backend

Medical Research Multi-Agent System 📄

- Orchestrated a 3-agent **LangGraph** pipeline (Retriever, Fact-Checker, Synthesizer) querying PubMed and arXiv in real time, embedding results into **ChromaDB** for semantic retrieval, and returning evidence-graded answers with citations
- Instrumented full agent tracing via **LangSmith**; surfaced results through a streaming **FastAPI** backend and **Next.js 15** + **TypeScript** frontend